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Success factors and strategies for sustainability transitions of small-scale communities – Evidence from a cross-case analysis

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ABSTRACT

Initiatives pursuing sustainability transitions of small-scale communities offer rich learning opportunities on: to what extent they are making progress toward sustainability; what success depends on; and how success can be replicated in other communities. Yet, there is little empirical research addressing these questions. This article presents a cross-case analysis of sustainability transition initiatives in four small U.K. communities that tentatively identifies factors influencing transition outcomes as well as strategies to improve transition effectiveness. Our findings suggest a wide range of factors to be important including socio-demographics, community governance, community capacity, organization and management, resources, and mobilization. The article demonstrates how empirical studies of this nature can successively build a broad knowledge base to support effective community transition practices.

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1. Introduction

There is growing interest in community-based sustainability transitions and how they can be accelerated, scaled, and otherwise improved (ARTS, 2014; Forrest and Wiek, 2014; Nevens et al., 2013; Seyfang and Longhurst, 2014). It is widely postulated that such initiatives at both city-region and small community scale have critical roles to play in societal sustainability transitions. To this end, a

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number of projects are now attempting to understand the mechanisms within these initiatives, identify critical success factors, and design and test strategies (Gupta, 2012; Seyfang and Longhurst, 2014; TESS, 2014; Wittmayer et al., 2013). Forrest and Wiek (2014) suggest such a task be approached by the accumulation of many case-studies from which robust solutions may be developed in the same vein as Ostrom's (2009) work on community-based resource management. In this article, we contribute to such an effort by presenting a cross-case analysis of sustainability transition initiatives in four small U.K. communities.

Small community sustainability transition initiatives adopt diverse approaches and settings (rural/urban; village, neighborhood, town, etc.), but they share the general description of an organized, collective bundle of actions – stretching over several years or even decades – intended to transform a community to a sustainable state (Forrest and Wiek, 2014). Typically, initiatives implement multiple projects simultaneously, such as promoting household energy efficiency or establishing community gardens (Forrest and Wiek, 2014). While they may be considered as belonging to a broad movement, community initiatives range from unaligned, independent affairs to well organized networks and include groups such as Transition Towns (Hopkins, 2008), low carbon communities (Moloney et al., 2010), and eco-villages (Dawson, 2006). There are perhaps a thousand or more active community initiatives, most commonly in post-industrial countries (Forrest and Wiek, 2014), with 472 active Transition Towns alone in November 2014 ("Official Initiatives by Number," 2014).

Such initiatives seek new ways of meeting the community's needs for food, mobility, housing, work, leisure and so forth that do not threaten its, or other communities', long-term viability and integrity (Forrest and Wiek, 2014). In socio-technical studies, sustainability transitions are conceptualized as a process of fundamental change in societal systems to more sustainable modes of production and consumption (Markard et al., 2012) beginning with innovation in protected spaces (niches) which may cause changes to mainstream institutions (regimes) and lead to widespread establishment of the innovation (Geels, 2002). In this context small community initiatives may be interpreted as 'grassroots' niches from which new social practices emerge and subsequently lead to mainstream societal (regime level) change at regional or national scale through replication, scaling-up, and translation of particular novel practices (Seyfang and Haxeltine, 2012). Thus, through various bidirectional mechanisms of interaction, community initiatives may contribute to transitions of urban or regional systems of provision such as electricity, waste management and housing (Seyfang and Haxeltine, 2012; Smith, 2007). However, community initiatives may be seen not just as sites of social innovation that contribute to particular socio-technical system transition at wider geographical scales but as in situ, comprehensive transitions of the community itself that cut across multiple systems of provision. Transition Towns, for example, see their primary purpose as transforming their own community although also acknowledging their contribution to a wider transition through their activities and membership of the Transitions Network. Thus, community initiatives are transitional in two respects: as in situ processes of transformation of the community itself; and as sites of social innovation or experimentation as part of regional scale transitions.

While both purposes are important it is to the primary purpose of in situ community transition that we turn our attention in this article and particularly to the question of how community initiatives can be more effective in this respect. This is a key question posed by Forrest and Wiek (2014) who propose a framework for its empirical investigation based on three research questions: what was done; was it successful; and how was it done. Studies using this approach can contribute directly to practical solutions in the form of guidelines for successful initiatives and contribute knowledge toward theory building.

The purpose of the present study is to identify factors that may be critical to the success of small community initiatives as sustainability transition processes. A multiple case study approach is used to investigate four small community initiatives in the U.K: Ashton Hayes, a small village in Cheshire, England; BedZED, an urban housing complex in London; Forres, a small Scottish town; and the Isle of Eigg, a remote Scottish island. The analytical–evaluative framework proposed by Forrest and Wiek (2014) is used to identify initiative outputs, assess outcome sustainability, and reconstruct processes for each of the four cases; as well as derive more general success factors through cross-case comparison. Three of the case studies were originally described in Forrest (2011) whereas this article focuses on the cross-case analysis and the research question of what factors influence the sustainability outcomes

Table 1
The four small community sustainability transition initiatives included in this study.

Name and location	Population	Transition		
		Type	Began	Description
Ashton Hayes, a semi-rural, commuter village in Cheshire, England	919	Going Carbon Neutral	2005	Community group led, aiming to become carbon neutral. Actions areas include: energy use; tree plantation; public transport access; community shop ("Ashton Hayes Going Carbon Neutral," n.d.; Ashton Hayes Parish Council, 2009).
BedZED, a mixed tenancy, urban housing complex in Hackbridge, London	220	Planned Development	2002	A private–non-profit professional partnership to stimulate further community sustainability. Action areas include: community garden; grocery delivery; car club ("BedZED," n.d.).
Forres, a market town in Moray, Scotland	9500	Transition Town	2007	Community group led, aiming to increase local resilience. Action areas include: community garden; farmers' market; energy use; plastic bag elimination; local food guide (Piper and Villani, 2009; "Transition Town Forres," n.d.).
Isle of Eigg, a rural island in Highland, Scotland	96	Community Trust	2007	Community group and governance led, aiming to reduce carbon emissions and increase sustainability. Action areas include: community gardening; energy use; transport ("The Isle of Eigg," n.d.).

of transition initiatives. A further step is taken to suggest strategies for practitioners that may make initiatives more effective.

The article proceeds as follows: Section 2 describes the research method; Section 3 briefly summarizes the results of each case study; Section 4 compares results across cases; Section 5 describes factors and strategies for successful initiatives identified from the cross-case comparison; and Section 6 concludes with some general discussion.

2. Method

The study consists of three phases: case selection; case studies; cross-case analysis.

2.1. Case selection

Cases were selected according to general criteria that they: (1) be place-based; (2) are "small" (population less than 10,000); (3) be community-based (they seek to change and are implemented by the community); (4) have sustainability as a primary goal. Further, only well-established initiatives with an apparently successful track record were considered on the premise that more can be learned from these than from examining failures. Additionally, a variety of transition types and settings was included. Table 1 briefly describes the selected cases.

2.2. Case studies

Individual case studies were performed using the evaluative–analytical framework developed by Forrest and Wiek (2014) for small-scale community sustainability transitions. The framework consists of three main parts answering the questions:

1. *What was done?* Identify the interventions (projects) implemented, outputs produced, and impacts on the community. Impacts are categorized as either structural (supply side) or behavioral (demand side) changes to the community.
2. *Was it successful?* Appraise the sustainability of the community impacts. Appraisal consists of a qualitative rating of an intervention's impacts against a set of established sustainability criteria (Gibson, 2006):
   - *Environment* – Establish and maintain long-term integrity of socio-biophysical systems.
   - *Wellbeing* – Deny no-one the means for a decent life or opportunity to improve.
   - *Equity* – Reduce gaps in sufficiency and opportunity between rich and poor to tolerable levels and protect opportunities and capabilities of future generations.
   - *Resources* – Reverse socio-ecological damage by doing more with less.
   - *Citizenship* – Prioritize openness and participation in governance and instill sustainability as a core value in all citizens and organizations.
   - *Adaptability* – Avoid risks of serious or irreversible damage to environment, wellbeing, or equity through anticipatory governance.
3. *How was it done?* Reconstruct the sequence of actions and circumstances leading to each intervention output. We use a method of reverse "process-tracing" (George and Bennett, 2005) to tease out intermediate and initial factors that appear to be important for the production of each output.

Data collection included visits to all sites in January 2011. Two semi-structured interviews of approximately two hours each were conducted at each site with interviewees purposively selected for their knowledge and experience of the initiative. Further data were collected from organizational, government and media documents including plans, reports, newsletters, websites, agendas and minutes.

2.3. Cross-case analysis

The objective of the cross-case analysis is to find factors that are important for successful transition processes i.e. that produce sustainable intervention outcomes.

We selected two interventions from each case for cross-case comparison. First, we selected similar interventions that were implemented in more than one case for the possibility therein to compare how differences in process led to varying degrees of success. There were two of these – household energy use and gardening for food – each implemented in three cases. Our other strategy was to select successful interventions of which there were no similar interventions in other cases. The rational for this strategy is to provide additional opportunity to identify common processes and enabling factors that lead to successful outcomes, even though the specific application differs. We also refer to the startup phase of initiatives which, of course, all cases have, to gain insights into the overall initiative-level process.

3. Case study results

In this section we present a brief summary of what was done, how it was done, and how successful it was from each case study. Fuller case study descriptions are available from the first author.

Common to all cases was the generally favorable political climate in the U.K. during the period examined toward "grassroots" type of action and in particular with respect to climate change and sustainability (Alexander et al., 2007; Amin et al., 2003; Seyfang and Smith, 2007). This manifested itself through various funding programs including the Scottish Executive's Climate Challenge Fund (Scottish Executive, 2010) and U.K. government lottery funding (NESTA, n.d.), from which Forres and Eigg both benefited.

3.1. Ashton Hayes

Ashton Hayes is a middle-class, well educated, commuter village of 360 households and population 916 in Cheshire, England. There is one primary school, a sports field and clubhouse, a small convenience shop, a village hall, one pub, and two churches. There are somewhat infrequent and slow local rail and

bus services but the predominant mode of transport is car and almost every household has at least one car. It is socially well-knit with several active community groups including the Women's Institute, Playing Field Association, and football club. Formal community-level governance is provided by an active parish council.[1]

The transition initiative began in 2005 and continues to the present day. Our analysis covers the period from 2005 to 2010.

The idea of a carbon neutral village was conceived in 2005 by a community member concerned about climate change and, aided by several acquaintances, the Going Carbon Neutral (GCN) initiative took off following a well attended and publicized public meeting in January 2006. The meeting and demonstration of community support had been a condition for the parish council's endorsement of the initiative. The general management approach is informal and adaptable, loosely driven by the ultimate goal (carbon neutrality) and a set of guiding principles, avoiding "an organizational structure that would slow us down" (Charnock and Alexander, 2007). There is no blueprint; opportunities are pursued as they arise. The guiding principles include that the initiative be non-political, non-confrontational, and non-judgmental. There is an emphasis on "the benefits of taking action" and not on the negative consequences of not (Charnock and Alexander, 2007). While officially an appendage of the parish council, the core management group of volunteers operates autonomously though significant decisions are ratified by the parish council and, in some cases, by public meeting. Volunteer workgroups are formed as needed to support the core group with specific tasks. Cash sponsorship (£3500) and in-kind support from local businesses has been helpful. Professional skills of community members, particularly media relations, marketing, and social science research, are utilized where possible. It is very much a community driven and controlled affair but there is no hesitance to overcome community capacity limitations by partnering with outside organizations including the local university, energy conservation non-profit, and local county council. Communication, both internal and external, through a variety of media, has been a prominent feature.

The public meeting created broad public support that was quickly transformed into participation in the *Household Energy Use* intervention. Villagers were asked to do whatever they could to reduce their energy use and increase their energy from renewable sources. The intervention focused on filling knowledge and skill gaps. Volunteers, organized into work-groups, worked with partner organizations to deliver "carbon clinics", utility program promotions, product and vendor information, and an annual carbon footprint survey and results feedback. Many actions were social, taking place in homes or the pub and engaging with community groups. Meanwhile, other interventions were pursued including *Tree Planting* (local landowners created plantations as carbon offsets); *Station Footpath* (providing pedestrian access to the village station); and *Household Recycling* (improvements to the council recycling service). In addition to Household Energy Use, however, the other intervention included in our cross-case analysis is the *Community Shop.* This intervention built on the community's capacity for action that had grown at least partly as a result of the GCN initiative. Alarmed by the impending closure of the only village shop, GCN collaborated with other community groups to take it over as a community enterprise. Following the deliberation toward making the decision and establishing a team of community members this was a relatively straightforward project to create a business plan, negotiate the transaction, and so forth. The community impacts of both interventions are shown in Table 2.

The initiative has produced some movement toward a sustainable community. The Household Energy Use and Community Shop interventions have contributed most to this. Although the greatest gains have been environmental (energy and CO_2 reductions) there have also been significant social and economic outcomes including increased community cohesion, engagement and, capabilities, and equity.

[1] Parish councils are the lowest level of local government in England. Ashton Hayes Parish Council serves the residents of Ashton Hayes and has direct dealings with the county level government (Cheshire County Council) on matters relating to the village.

Table 2
Summary of community impacts of the two Ashton Hayes interventions included in the cross-case analysis.

Domain	Infrastructure/supply side	Activity/demand side
Household energy	• No or low cost (with rebates) building upgrades such as replacing light bulbs or shower heads, or insulating walls. • Higher cost building upgrades such as new windows. • Higher cost energy conversions including solar water heaters, solar PV, and wood burning stoves. • Replacing old cars with more fuel efficient cars.	• Behavioral changes such as turning off lights when not in use or adjusting thermostats. • Villagers have reduced the number of flights by 37% though much of this has been employment related. • Awareness of energy use has become part of the village culture. • Increased social interaction.
Community shop	• Created a community owned and operated enterprise. • Avoided loss and enhanced access to and level of basic goods provision. • Increased involvement of the community in providing for their own needs. • Created a new outlet for local producers. • Created a revenue stream for community development. • Maintained and enhanced public. space. • Extended community communication infrastructure (notice board).	• Three community members work in the shop. • Several community members regularly volunteer in the shop. • Most community members make fewer and shorter grocery shopping car trips. • Community members shop locally more than before. • Increased social interaction between community members. • Increased awareness of community activity among community members.

3.2. BedZED

BedZED is a sustainable housing and office development in Hackbridge, in the London Borough of Sutton. There are approximately 220 residents in 99 mixed tenure dwellings (private, partially owned, and local authority rentals). It is well served by bus and train and car ownership is lower (59%) than Hackbridge (84%) although for owner occupied households it is about the same (Hodge and Haltrecht, 2009). The property is managed by Peabody, a large social housing association and limited community governance is provided through a resident's association. Although there is little organized community activity, informal social cohesion among residents is strong.

The project was developed by a partnership between Bill Dunster Architects, BioRegional, a non-profit sustainable development consultancy, and Peabody, the main funders, from 1996 to 2002. In many respects it was a traditional construction project but unlike purely commercial projects it also had social purpose: to create a sustainable, urban community "attractive and affordable" to all (BRESCU, 2002). Achieving the aim is premised on a three-tier design ("BedZED," n.d.). Tier 1 is technical (low building ecological footprint), tier 2 is behavioral (steer residents toward sustainable choices through design), and tier 3 is social (tiers 1 and 2 create a strong community that pursues further sustainable development). There was no U.K. policy support for sustainable housing development at that time (Lovell, 2008), but Sutton Council played an important role in facilitating the project (Minton, 2001). Future residents were not involved in the development although local residents were consulted.

Our analysis begins on completion of construction in 2002. At this point, a foundation had been created providing intrinsically strong sustainability outcomes for any resident, particularly from low energy and water use, low utility bills, equitable housing, and good transport options. Furthermore, the potential was there for a strong and sustainable community with a high quality of life to develop. To stimulate this potential a Green Lifestyle Officer (GLO) was funded for the first year through a government SEED grant while BioRegional and BDA staff engaged with residents in regular, often sustainability themed, social events. Three specific interventions were introduced: the *Community Garden*, *Car Club* and *Grocery Delivery* of which the first two are included in this cross-case analysis. For the Community Garden BioRegional secured a land lease from Sutton Council and, through public funding, installed facilities and provided tools and resources. The GLO provided encouragement, information, training, and resources to residents individually and in workshops, and introduced them

Table 3
Summary of community impacts of the two BedZED interventions included in the cross-case analysis.

Domain	Infrastructure/supply side	Activity/demand side
Community garden	• Lease from local council on land. • Raised beds/allotments and infrastructure. • Management by GLO. • Membership free to all residents. • Shared tools and resources provided.	• Gardening was initially popular but declined after the first year. • Some composting. • Popular informal social space.
Car club	• Onsite location. • 3 cars initially, later expanded. • Membership offered to BedZED residents and businesses at reasonable cost.	• 50 members in 2004, 31 in 2009. • 9 household (10%) members. • Low car ownership (0.6 per HH). • Low car miles (2300 per HH).

to local suppliers. The Car Club was a partnership with a commercial operator offering membership to BedZED residents and businesses and later expanding to Hackbridge residents. Membership was encouraged through parking restrictions and active marketing by the GLO who also worked to improve the service. Community impacts for these interventions are described in Table 3.

BedZED has made substantial progress toward sustainability. Residents have a markedly lower ecological footprint and apparently higher quality of life than in neighboring communities (Hodge and Haltrecht, 2009). However, these outcomes are mostly attributable to the foundational aspects of the community, namely, tiers' 1 and 2 physical and institutional design and implementation. The interventions delivered from this base have produced less convincing outcomes. The car club, while economically viable and supporting a small section of the population to lead low car-dependent lives, has not substantially changed resident mobility and the community garden, despite some initial success, has produced very marginal sustainability outcomes in social cohesion (as an informal gathering place) and waste reduction (composting).

3.3. Forres

Forres is a Scottish market town whose 9500 people are somewhat more demographically diverse than Scotland generally with relatively large English and European populations (General Register Office for Scotland, n.d.). This reflects the influence of the nearby airforce base and Findhorn Foundation. The population is generally well educated with a broad mix of employment. It is served by rail and bus services and 28% of the 3800 households have no car, though the predominant mode of commuting is by car (General Register Office for Scotland, n.d.). There is an active high street, a national supermarket, a high school, several primary schools, and a private higher education college. The voluntary sector is active with a large number of community groups.

The Forres transition initiative began in 2008 after its three founders, who met through the Findhorn Foundation, decided to apply the Transition Town concept in Forres. Its goal closely aligns with the movement's goal of building "local resilience in the face of the potentially damaging effects of Peak Oil and Climate Change" (Piper and Villani, 2009). The initiative generally follows the Transition Town approach (Hopkins, 2008). Transition Town Forres (TTF) is incorporated as charitable company and has a formal organizational structure comprising a board of directors, a management committee, and multiple project teams. A three-year plan (2009–2012) sets out the initiative's objectives. There is a strong emphasis on professional, business-like management where, as a leading member put it, core management members are asked to "both put the spade in the ground and . . .to write up minutes from meetings". Although responsibility for projects is delegated to workgroups led by a project manager, they are accountable to the management committee for decision making, reporting progress, and justifying budget requests. TTF is primarily funded through grants including £184,000 from the Scottish Government in 2008. TTF depends heavily on volunteers and has cultivated strong, mutually supportive links with other community groups but has also used funding for staffing and consultancy.

Five interventions were implemented in the three year plan: a *Plastic Bag Elimination* campaign; establishing a *Community Garden*; establishing a *Farmers' Market*; publishing a *Local Food Guide*; and

Table 4
Summary of community impacts of the two Forres interventions included in the cross-case analysis.

Domain	Infrastructure/supply side	Activity/demand side
Household energy	No Impact	• Pledges by ~4% of households which could reduce carbon emissions by up to 20% per household. • Pledges are mostly home energy conservation actions such as adjusting thermostats or air-drying clothes. • Other pledges come from mobility, food choices, and consumer goods purchases.
Community garden	• 11-year lease from local council on 1.5 acres of common ground. • 14 permaculture style pods. • Autonomous management by members. • Permaculture based rules (no artificial fertilizers or pesticides). • Membership open to all residents for annual fee of £40. • Shared tools and resources including a tractor, worm bed composting, bee keeping. • A new community space.	• Fully subscribed by 71 gardeners. • Substantial food produced for personal consumption (no data on exact quantity). • Gardeners share experiences and learn from each other. • Enjoyed by gardeners as social activity ("a friendly and open space"). • Increased interaction within the community and raised awareness of community and transition activities.

Household Energy Use reduction. The Community Garden and Household Energy Use interventions are included in the cross-case analysis. Obtaining the land for the community garden entailed difficult and protracted negotiation with a generally unhelpful local authority. Meanwhile, within the broad parameters set by the management committee a volunteer workgroup created detailed plans, publicized the development, purchased equipment, prepared the site, and recruited members. Now operational, the workgroup manages maintenance and membership, provides an open forum, and holds expert workshops. Additionally, other transition activities such as local food cooking classes create wider interest. In the Household Energy intervention 160 households were recruited to participate in an energy use survey and, based on the survey results, pledge behavioral changes, which are counted as carbon emission reductions. A follow up phone call 12 months later reviews adherence to the pledges. The intervention was initially performed by a knowledgeable volunteer but later contracted to a consultancy. Community impacts for these interventions are described in Table 4.

The Forres initiative has had a substantial impact in terms of raising community awareness, engagement, and participation, especially so through the extensive reach of the farmer's market and local food interventions. Concrete sustainability outcomes, however, are more difficult to claim, especially at the town level. The most important contribution of the community garden is its social cohesion, equity, and governance aspects and, less tangibly, the possibility that it has created a site in which alternative ideas can grow and diffuse into the wider community. The Household Energy Use intervention contributes mostly to environmental sustainability with perhaps up to 20% CO_2 emission reductions. However, this is in only 4% of households and, given the unknown effectiveness of pledging, the overall impact may be quite small.

3.4. Isle of Eigg

The Isle of Eigg is a small, remote Scottish island with a population of around 90 comprising indigenous islanders and incomers. Its economy is based on agriculture, tourism, and freelance work. The island has one single-track road and a mini-bus service on ferry days (three days per week). There is one shop, a tearoom, a primary school, a community hall and a church. Almost one in two adults volunteer in at least one of over 20 community groups (Bryan, 2007). The island (land, infrastructure, commercial buildings and much of the housing stock) is owned by a community land trust. A board (50% community and 50% external representation) makes strategic and operational decisions but takes ultimate direction from the Residents' Association. An executive group, composed of a fulltime administrator and four voluntary directors, carries out the board's directives.

Table 5
Summary of community impacts of the two Eigg interventions included in the cross-case analysis.

Domain	Infrastructure/supply side	Activity/demand side
Household energy	• Energy efficient appliance upgrades by most households. • Energy efficient building upgrades, e.g. (insulation, new doors) of several houses. • Solar water heaters installed on several homes. • Heating fuel conversions from coal to wood.	• Behavioral changes such as turning off lights. • Increased awareness and knowledge of energy use and impacts (although already relatively high). • Sharing and learning from each other.
Community garden	• At least 7 households erected polytunnels. • Permaculture methods encouraged (no artificial fertilizers or pesticides). • Free club membership open to all residents. • All households have composting equipment.	• Many islanders now grow food. • Substantial food produced for personal consumption (no data on exact quantity). • gardeners share experiences and learn from each other. • Many islanders actively compost.

The community has undergone a radical transformation over the last 20 years from an underdeveloped and declining state after a long and contentious buyout from oppressive external landlords in 1997 (McIntosh, 2001). Since then a process of continuous, community-driven development has made major quality of life, economic, and ecological improvements (Bryan, 2007). Our focus begins in 2007 with a turn toward smaller scale, grassroots development when the community undertook a review process and began to more explicitly expose its underlying sustainability values through the "Green Island" concept ("About Green Eigg," n.d.). The ambitious transition initiative of the Isle of Eigg has received media attention around the world.

Eigg's general transition approach consists of a core group of volunteers (the "Green Team") operating within the established governance structure of the Trust. This arrangement affords administrative support, formal institutional status, accountability, and legitimacy to the Green Team while ensuring integration of transition activities with other community activities. Any community member can volunteer for the Green Team, but it has mostly been driven by a few individuals, some of whom have backgrounds in community development and environmental action. Later in 2007, the island submitted a plan to a U.K. Lottery funded competition for community-based climate change action, which led to the award of £20,000 to execute the plan and a further £138,000 from the Scottish Executive for associated projects. The goal was to reduce carbon emissions as much as possible in a 12-month period. Four interventions were implemented: *Road Improvements* to increase walkability and cyclability; a *Gardening Club* to increase food production; *Mobility Options* to increase transport alternatives; and *Household Energy Use* to reduce energy use and carbon emissions. We include the Gardening Club and Household Energy Use interventions in our cross-case analysis.

In the Household Energy Use intervention the Green Team were supported by the University of East Anglia's CRed project to monitor and analyze the community's carbon footprint. Islanders were offered small grants for energy efficient appliance upgrades, subsidies for building improvements and energy generation, practical advice and training. There was continuous mobilization of the community through intense interaction on multiple levels. Learning occurred through feedback of results and sharing each other's experiences in formal groups and informal interaction. The Community Gardening intervention consisted of establishing a gardening "club" to "share information, advice, seeds and equipment" (www.islandsgoinggreen.org); providing compost bins and expert training to every house; providing grants to purchase polytunnels; and holding an annual local food community event. Community impacts for these interventions are described in Table 5.

The two interventions described here produced successful sustainability outcomes through both effective delivery and wide reach. The Household Energy intervention not only reduced CO_2 emissions by ca. 28%, but also did so equitably whilst also increasing the community's strength and capabilities. The Gardening intervention has also been equitably beneficial, whilst also contributing to a more resilient community.

4. Results from cross-case analysis

We look for patterns and generalizations by comparing similar interventions (*gardening for food*, and *household energy*) across the cases. To gain further possible insight into more generalized success factors we also draw on the *Community Shop* and *Car Club* as relatively successful interventions from Ashton Hayes and BedZED respectively but with no direct counterpart in other cases, and from the startup stages of each case. We combine insights gleaned from the individual case studies as to what appeared to produce more or less successful outcomes and whether such factors had a similar influence in other cases.

As a starting point, an overview of the sustainability appraisal of each of the eight interventions included in the analysis is presented in Fig. 1. While all interventions had some positive impact, there are clearly some differences between similar interventions and between different interventions within the same initiative. Sustainability appraisal reflects both the effectiveness of the intervention and its reach across the community.

4.1. Household energy

Household energy interventions were carried out in Ashton Hayes, Forres and Eigg with goals of reducing household level carbon emissions.[2] While all achieved success, community level impact and longevity of effect appear to be strongest in Ashton Hayes and Eigg.

Ashton Hayes's main intervention activities were: providing information (practical advice, accessing resources, performance feedback) and monitoring performance (annual survey), presented as a collective challenge (carbon neutral village). Activities were delivered at household level (individualized data, personal contact), community level (broad communication and workshops) and sub-community level (group events). This socialization appears to have encouraged social learning and innovation in which new solutions and social norms emerged that have been important for 20% energy use reduction sustained over four years (Alexander et al., 2010).

Eigg's activities were similar to Ashton Hayes – providing information and monitoring performance – but also included providing resources (grants). Other similarities were collective framing of the intervention (to maximize community carbon emissions reduction) and multi-level delivery to households (individualized and personal), groups, and community. Average direct household carbon emissions were reduced by up to 20%.[3]

The Forres intervention consisted of monitoring (self-reported assessment and pledge) and providing information (feedback and tailored advice) but differed from Ashton Hayes and Eigg in being primarily household level delivery with contact limited to an initial in-person consultation and a planned 12-month follow up phone call. In comparison, ongoing activities, including monitoring, were conducted over four years in Ashton Hayes and twelve months in Eigg. Although Forres results indicate carbon emission reductions comparable to Ashton Hayes and Eigg, participation was much lower (four percent of households) and it is unknown if participant pledges were fully enacted and subsequently sustained.

While all three interventions appear to have successfully reduced carbon emissions among participants, Ashton Hayes and Eigg were more successful in reaching the whole community than Forres and, possibly also, in achieving sustained change. This is based on the uncertainty that participant pledges in Forres were followed through and adhered to, whereas change actually happened and was lasting in Ashton Hayes and Eigg. Although pledges can be effective when combined with feedback and other strategies (Bartram, 2009), follow-up studies often show that intervention effects are short-lived (Abrahamse et al., 2005). To explain this, we point to the nature of the Ashton Hayes and Eigg intervention as being multi-stranded and largely social, involving frequent interaction with and between

[2] Household energy use is defined here as direct energy used in and around the home for heating, cooling, appliances, lighting and such like. It does not include transport or indirect energy use through consumption of food or other goods.

[3] Data were not obtained for direct household energy but as per note 2 total emissions were reduced by 28% of which transport and consumption are assumed to be relatively small contributions due to their initially low values.

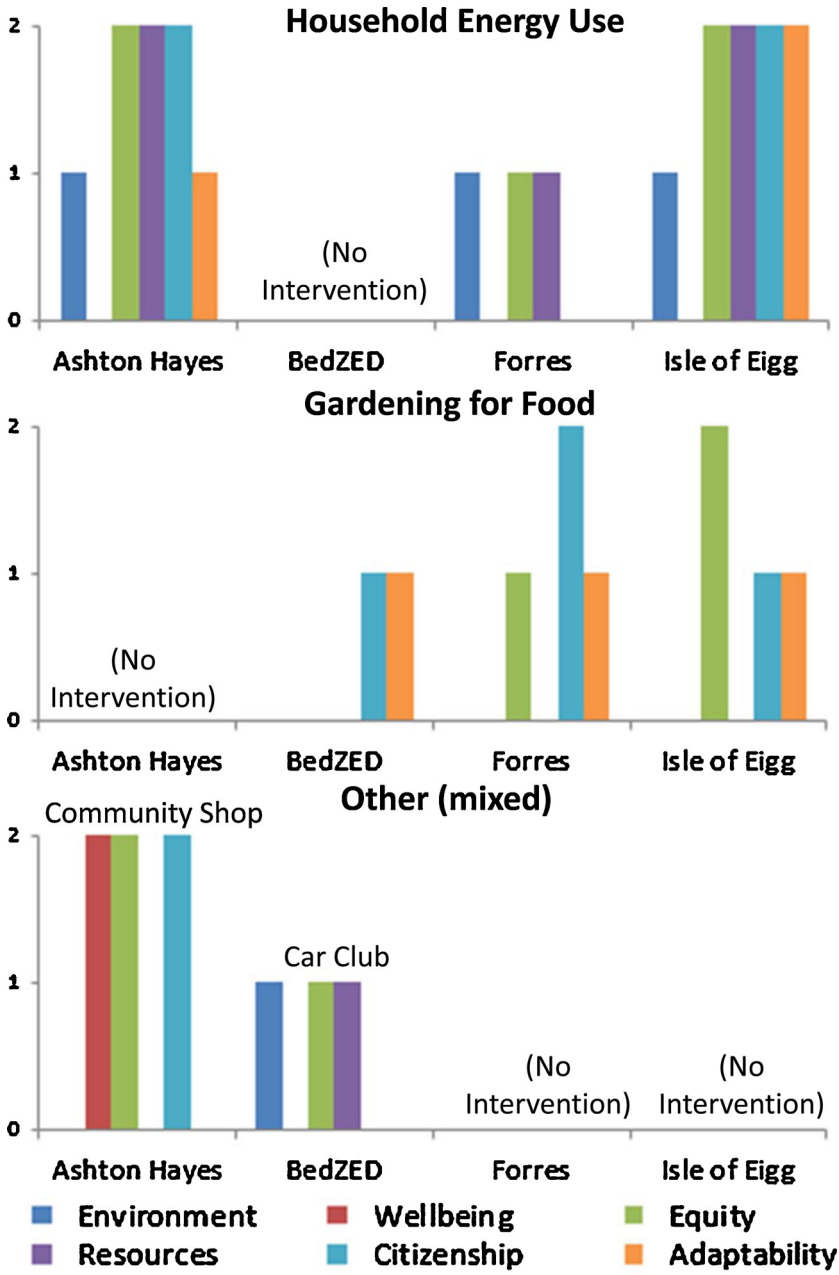

Fig. 1. Overview of sustainability appraisal results included in the cross-case analysis. Each intervention is rated against six sustainability criteria where 1 = marginal change, 2 = substantial change, and >2 = very substantial change. Interventions from each initiative are stacked vertically while the top row is the Household Energy Use intervention (no intervention for BedZED), middle row is gardening for food (no intervention for Ashton Hayes), and the bottom row is mixed.

participants. Abrahamse et al. (2007) confirm that individual feedback, group feedback, and group forums can be particularly successful strategies. Energy use in these communities became a central topic of conversation. In Forres, the intervention was mainly limited to a single action (feedback and tailored information) delivered to isolated householders dispersed throughout the town, who may not have known any other participants. This missing socialization process prevents the emergence of mediators of social learning (Darby, 2006) and new social norms (Abrahamse et al., 2005). These are important for the diffusion of social innovations and can strengthen psychological determinants of sustained lower energy use. Ashton Hayes and Eigg, however, have an advantage over Forres regarding intervention socialization in that they are much smaller communities: population size and existing social cohesion would appear to be critical factors that enabled these social mediators. We suggest then that group activities are especially important in settings where inter-participant interaction is less likely to occur, such as in larger communities with dispersed participation like Forres.

Another factor is goal setting, which can be important in energy use interventions (Abrahamse et al., 2005). There were differences in the framing of the intervention objectives as more individual or collective. Ashton Hayes and Eigg appealed to community members to contribute to the greater cause while potentially benefiting individually too. The goals in Forres were primarily targeted at households to reduce their carbon footprint set against somewhat vague and abstract transition town goals of building local resilience. The tangible, though ambitious, community goals set in Ashton Hayes and Eigg may have exerted pressure on community members to participate for the good of the community. This is an effect that would be reasonably expected to be moderated by existing social cohesion and also if the goals were widely agreed through established community governance mechanisms. While Transition Town Forres has substantial support in the town and follows open, transparent decision making, it cannot claim to have a mandate to act on behalf of the community to the extent that Ashton Hayes and Eigg can. Neither does Forres, due to its larger and perhaps also slightly factional population, have the same strength of social cohesion as Ashton Hayes or Eigg. Socio-demography is quite probably important in this, particularly population but possibly also homogeneity, education, and other aspects.

4.2. Gardening for food

BedZED, Forres, and Eigg all implemented interventions with the goal of establishing or strengthening a practice of growing food in the community. More distal outcomes of reduced carbon emissions, lower ecological footprint, greater self-sufficiency, and nutritional health were also hoped for. We focus here on the proximal outcome (a practice of food growing) for which Forres and Eigg achieved considerable success, but BedZED less so despite a promising start.

The Forres intervention consisted of: providing resources (land, equipment and tools); providing education (practical and cognitive knowledge); setting standards (adopting permaculture rules); and establishing governance (gardener autonomy). These ingredients reframed gardening from a relatively isolated, private pursuit to one of a positive, social and learning experience. Several factors contributed to this: the garden site itself brings people together and generates social interaction; the autonomous governance encourages participation and requires cooperation; and the permaculture layout of plots may induce social integration. This breaking down of individual barriers promotes skills and experience sharing, creates a mutually supportive environment, and develops collective responsibility and cohesion amongst gardeners. Another meaning constructed is that of gardening as a holistic, natural process, achieved by the application of permaculture rules and the introduction of specific material elements (e.g. compost wormery, bees).

Eigg had similar activities to Forres: providing resources (compost bins, polytunnels, equipment and tools); provided education (organized training, sharing of skills and experience); setting standards (permaculture emphasis); and establishing governance (self-managed club). We suggest that these activities have strengthened particular meanings associated with gardening, specifically its natural and holistic nature and it being a logical part of household production. This framing worked well on Eigg where basic resources (land and water) are abundant and household livelihoods typically depend on multiple sources of income. Gardening on Eigg, however, is not the same type of social experience that it is in Forres as it happens on private plots and not on a communal site. Socialization of the

intervention, however, was still achieved by means of the "gardening club" but also was helped by the existing strong social cohesion.

BedZED's gardening intervention failed to take root and today little gardening is done in the allotments. Like Forres and Eigg, resources (land, facilities, tools, seeds, etc.) and education (training) were provided. Leadership and organization was provided by the Green Lifestyle Officer (GLO) but after early success, participation declined when the GLO position ended. The Forres and Eigg gardening communities are self-governing whereas in BedZED, organization and management was left to the GLO, an outsider. BedZED gardeners, then, appear not to have developed sufficient interest and perhaps competencies required to sustain the practice. Perhaps the intervention failed to create new meanings strong enough to sustain the practice or to develop levels of competency needed to confirm the benefits of the practice.

Another difference is in the transition context. In Forres and Eigg the gardening interventions were delivered within an organized transition initiative with structure, goals, and plans. This provides benefits that are more difficult for isolated interventions to attain, including: legitimacy, credibility, and accountability; communicating with the community; sustaining participation; administrative support; creating greater meaning beyond gardening only; and setting collective goals. BedZED, however, had no transition initiative. It is suggested then, that strong transition initiatives are an important factor in the success of individual interventions: without the wider support that comes from being part of larger endeavor it is more difficult for individual interventions to take root and grow. More is said about this below.

4.3. BedZED: the missing transition

In comparing community gardening interventions above the importance of the transition context was recognized in providing support for interventions to take root and grow. Three of the cases – Ashton Hayes, Forres, and Eigg – had much stronger transition contexts than BedZED and, in terms of providing the support necessary for effective sustainability interventions, these three cases were more successful, based on the number and outcomes of interventions attempted. While BedZED is a different type of community transition from the others and a simple comparison is therefore not entirely valid, it still exemplifies what happens when there is no organized, grassroots, purposeful initiative in existence. Creating this type of initiative was not the primary goal of BedZED, but tier 3 of the BedZED design strategy does aim at the community creating "their own facilities and groups to improve quality of life and reduce their environmental impact" ("BedZED," n.d.). This tier-3 outcome, however, has not materialized and this, as suggested above, is one reason why the gardening intervention has not taken off to the same extent it has in Forres and Eigg.

In other respects, BedZED shares contextual characteristics, which were identified earlier as perhaps being significant in sustainability interventions. The small population and strong social cohesion creates ideal conditions for social innovation and diffusion as seen in Eigg and Ashton Hayes. Its mixed demography, consisting of low income, working class, and environmentally minded professional groups, has broad parallels with Eigg. Another similarity with Eigg is the strong infrastructural sustainability baseline, particularly energy, from which initiatives started. Also in common with Eigg, and with Ashton Hayes, is the huge amount of national publicity BedZED has received. In Eigg and Ashton Hayes, the publicity appears to have been a positive influence on mobilizing the community but apparently has had no influence on BedZED since the first year or two after residents moved in. This is partly due to BioRegional deliberately, and quite understandably, protecting the community's privacy. Overall, these factors do not appear to result in sustainable development efforts unless harnessed by an organized, grassroots sustainability transition initiative.

The lack of a BedZED grassroots transition initiative has not prevented the car club from being successful. We attribute this to the car club being a fundamentally different type of intervention based on the market economy, though with social purpose, and not just on community support. The car club is an integral part of the BedZED tier-2 design and although initiated by BioRegional and supported in its early development through public funding of the Green Lifestyle Officer (GLO), it is commercially owned and operated by a private business. Unlike the garden, the car club was able to sustain itself after withdrawal of the GLO because there was an organization in place with a vested

interest and high degree of responsibility for its continuation. Car clubs require specific conditions for viability – BedZED's owes much to business memberships from the co-located BioRegional office as well as to its urban setting with good access to public transport.[4] However, the general point is that interventions with a strong market base may be robust solutions even in the absence of strong community initiatives.

5. Success factors and strategies

We now identify some general factors that we suggest are important to successful transitions. These are categorized into contextual factors existing pre-initiative and intermediate factors that developed during the transition process. We finally highlight some strategies that appear to have contributed to these factors.

5.1. Contextual factors

Socio-demographics: Population size was pointed to as a primary factor affecting engagement and mobilization of the community. Forres (population ~10,000) had a much harder task than Ashton Hayes (~1000) and Eigg (~100) in this respect. Larger population makes effective communication more difficult, has looser social cohesion, and creates practical limitations to effective intervention delivery. This suggests there may be a population threshold somewhere between 1000 and 10,000, above which different strategies become necessary. Population, though, as demonstrated by BedZED, is not a simple, independent factor, but more likely a moderator when other factors are present.

Community governance: Existing community governance played a direct part in the transitions of Ashton Hayes and Eigg. In Ashton Hayes, the parish council was prepared to be a vehicle for the transition group, providing legitimacy and administrative support, on condition the community supported such an act. In Eigg, the Trust made the transition initiative an official community development project. In Forres, where there was no such relationship, the initiative had to build its own legitimacy and authority. The lack of such a relationship is perhaps a contributing factor as to why Transition Town Forres is not as widely embedded in the community as the transition efforts in Ashton Hayes and Eigg. Additionally, however, it seems important that the Ashton Hayes and Eigg institutions taking in the initiatives were open, transparent, and trusted by their communities–which is not the case in every community. They were also somewhat progressive in their willingness to try new directions and projects if there is community support. Eigg had great scope in this respect given the extraordinary amount of control over development afforded by community ownership of the island.

Skills and experience: All three transitions depended to a considerable extent on skills and experience of volunteers. In Ashton Hayes this included media relations, social research, and film making expertise. In Forres, it included substantial business administration and community development experience. In Eigg, it included the extensive project administration experience of the Trust, as well as community development and sustainable development experience of volunteers. Additional expertise from project partners or professional consultants, however, was still necessary in all cases.

Funding opportunities: Public funding was important in Forres, Eigg, and BedZED. Without it, Forres may not have got beyond being a talking shop, and Eigg possibly would have no coherent sustainability initiative. BedZED's gardening success lasted only as long as the funding for the Green Lifestyle Officer. Ashton Hayes, however, demonstrates that significant achievements are possible without public funding and only very limited private sponsorship. Ultimately, initiatives must find ways to make interventions financially viable such as the BedZED car club and the Ashton Hayes community shop.

Community: Social cohesion was earlier identified as a possible factor leading to greater participation in Ashton Hayes and Eigg. Also important is an existing network of mutually supportive community organizations providing resource sharing and collaboration opportunities. In Ashton Hayes, for example, help was received from, among others, the Women's Institute and the primary

[4] In contrast, though not included in this study, Ashton Hayes's attempt to establish a community car club was discontinued after one year with lack of public transport and high car ownership cited as a reasons ("Nissan Leaf is now for sale", 2012).

school, and in Forres there were collaborations with numerous organizations including a health and wellbeing organization and a re-use center.

Core group and Leadership: A core group of volunteers and to a lesser extent individual leadership are important factors in driving transitions. Both Ashton Hayes and Forres initiatives were started by a single individual who worked closely with two or three others and who has remained as the recognized leader. In Ashton Hayes the core group has changed slightly, but is still the core management team. In Forres, the original core group gave way to the larger, formally structured management committee. In Eigg, individual leadership is perhaps more subtle, but the "Green Team" has been a critical core group.

5.2. Intermediate factors

Resources: Interventions depend on the availability of some set of resources. This includes materials (e.g. compost bins given to Eigg households), facilities (e.g. meeting rooms), land (e.g. for gardens), and labor (voluntary or paid). The availability of such resources is often a limiting factor in what can be done. Frequently there is a reliance on the goodwill of participants to use their private means but in all of the interventions analyzed this was insufficient. Transition organizations expended significant effort through, for example, grant applications, forming partnerships, and negotiating agreements – securing the resources needed for interventions.

Organization and management: The organizational structures of the Ashton Hayes, Forres, and Eigg initiatives were all operationally effective and share three common features. First, they are built around a formal, legally recognized entity: the parish council in Ashton Hayes (local government body); the Trust in Eigg (limited company); and Transition Town Forres in Forres (limited company). Each includes a board of directors providing oversight and guidance. This formal backbone provides: legal status necessary for conducting business transactions; accountability; and legitimacy in the eyes of much of the community. Second, is a management committee: a core team, usually voluntary, that makes operational decisions and performs other management tasks. Third, is the delegation of responsibility for intervention implementation to workgroups. Overall, this structure seems to provide a balance between formal responsibilities and operational flexibility that allows a diverse set of volunteers to find roles and responsibilities that fit their capacities and interests.

Community support and participation: Community support affords initiatives legitimacy even as far to claim that they are acting on the community's behalf. This was certainly so in Ashton Hayes where support was overwhelmingly demonstrated at the initial public meeting and in Eigg through the involvement of the residents' association in community decision making. Forres was somewhat different in that it was incrementally building support through awareness raising events and forging links throughout Forres's extensive network of community organizations. While the Forres transition could not claim to have full community endorsement, it has established substantial grassroots support across the wider community that provides significant legitimacy beyond that of the originating group.

Community support, no matter how widespread, does not automatically translate into participation. Ashton Hayes, Forres, and Eigg had varying degrees of success in this respect with numerous factors playing a part. Substantial positive media attention in Ashton Hayes and Eigg created excitement and enthusiasm, but the external interest may also have strengthened members' belief that their actions are worthwhile and increased members' desire to not to let the community down. A groundswell effect where people want to be part of a movement and peer pressure to conform may also have played a part. Rational decisions to maximize individual utility were likely a factor in Eigg where direct material support was provided. A history of successful community development such as on Eigg will likely predispose community members to participating in future projects.

We suggest also that salience of the transition goals to community members is an important motivation factor. Salience of the need to reduce carbon emissions in Ashton Hayes already seemed to exist: what was needed was positive encouragement to do something about it. On Eigg, salience also already existed from previously experiencing the benefits of community development and alignment of community values with sustainability. Furthermore, Ashton Hayes and Eigg set clear, long-term goals framed as challenges to the community that may have given added purpose and impetus to interventions resulting in increased participation. In Forres, it is not clear that the salience of the

Table 6
Tentative strategies for communities to build strong transition processes and implement successful interventions.

Transition strategies
Form a core group of committed activists.
Work with the established community governance if it exists, is trusted and accessible.
Create a formal organization and invite selected community members to sit on the board.
Find common ground and opportunities for collaboration with other community organizations.
Build community support and interest.
Adopt and maintain openness and transparency within the organization.
Set clear and simple long-term goals to motivate, direct, and allow progress to be measured.
Develop strategic partnerships to obtain necessary resources, services, and knowledge.
Frame the transition initiative in ways that are most relevant to the community.
Intervention strategies
Delegate responsibility to workgroups for interventions within broad transition level parameters.
Set collective, community level goals.
Socialize intervention activities wherever possible and in multiple ways.
Deliver intervention activities over an extended period.
Allow participant/volunteer flexibility and encourage creativity to stimulate innovation and social learning.
Monitor performance and provide regular feedback to participants at multiple levels.
Look for synergistic and multi-party collaboration opportunities.
Develop revenue generation or other economic value creation to achieve viability.

Transition Town objectives really penetrated and resonated with most of the population. However, it is clear that when Forres offered positive solutions in the form of specific interventions the population responded strongly. This is seen with the oversubscribed community garden, the well-attended farmers' market, and the rapid distribution of local food guides. In this case, salience may have less to do with "the potentially damaging effects of Peak Oil and Climate Change" and more to do with alignment of community values and interests.

5.3. Strategies

From the preceding cross-case analysis, some basic strategies that may help community practitioners build strong transitions and enact successful interventions are tentatively identified (Table 6).

These strategies are practical guidelines for the process, not the substance, of change. We differentiate between transition strategies which are concerned with steering change toward long-term goals and intervention strategies concerned with implementing discrete impulses of change. Transition strategies provide ongoing direction, continuity, organization, and resources and are typically performed by a management group. Intervention strategies guide design and delivery of individual projects and may include more specialist participants. Though not explored here, there are many possible links between strategies of either type. Thus, whilst enacting one or more strategies may provide marginal improvements in outcomes, greatest success is more likely when many or all of these strategies and others are enacted within a clearly established and organized community-based initiative.

6. Conclusions

The factors and strategies identified through this study are not intended as a comprehensive or definitive guide for successful initiatives. Yet, they illustrate the direction this research approach could take. The findings are tentative given limitations of the research and its exploratory nature. The limited data collection and somewhat crude sustainability appraisal method are methodological details that can be improved upon, for example. With the accumulation of findings from further similar studies, a more comprehensive set of factors can be constructed and subsequently tested in targeted case studies. While this study demonstrates the value of such an empirical approach, there is also strong complementarity with theory-driven approaches to transition initiative research. Although this study focuses on in situ sustainability transitions of communities there is also a need to link such community level

processes to wider societal transitions through grassroots innovation such as Seyfang and Hexaltine's (2012) study of the Transition Network.

Given the exploratory nature of the study, we did not attempt to differentiate strategies by context, but this is clearly important. Socio-demographic characteristics and social capital are important sources of internal influence that strategies should be sensitive to. Similarly, general setting and external factors such as funding availability or belonging to a network of transition initiatives may be critical factors. Another area we left largely unexplored was the extent to which the substantive nature of an intervention may be a factor in outcome success and how this may also relate to contextual factors. Are some types of community, for example, better suited to community gardens than energy use interventions? These are questions we suggest to be taken up by future studies.

While this article is focused on elucidating factors and strategies that might increase intervention effectiveness through improved processes, it does not squarely question the transformational potential of interventions. In other words, how far could an intervention possibly advance an in situ sustainability transition of the community? The sustainability appraisal results indicate that most interventions do produce gains in sustainability but they are marginal compared to the magnitude of the task. The most substantive gains seen were in household energy use. This, however, appears to be relatively easy pickings and it is not obvious where further gains will come from. Other domains such as mobility and consumption seem more intractable. Even with highly effective delivery, it appears that many interventions undertaken by initiatives would not be transformational. This is not surprising given the limited control of communities over their everyday practices and surroundings and the few resources at their disposal (Barr and Devine-Wright, 2012). On the other hand, the results indicate that interventions can strengthen social capital and community empowerment. Though less tangible than, for example, CO_2 emission reductions, such outcomes may in the long run be more important to enable communities to challenge the root political and economic causes of unsustainability and meaningfully change systems of provision (Chatterton and Cutler, 2008; North, 2009; Steffen, 2009). Community developments in Eigg prior to the activities analyzed here included truly transformational interventions in governance, property rights, housing, and energy; they may be exemplary of what is needed in this respect. We suggest evaluating transformational potential is a critical consideration that must be incorporated into the framework and used in further studies.

As initiatives navigate a pathway toward their objectives there may be dilemmas about how to proceed. Although not always occurring clearly as either/or choices in the cases studied here, they may be recognized as such in future encounters by other communities. For example, in all cases except Ashton Hayes, public funding was generally seen as the most accessible means of obtaining the resources needed to manage and implement projects. However, particularly in Forres and Eigg, applying for funding, managing funds, and reporting to funders was a significant burden that required a more managerial, professionalized approach and paid staff which in turn increased the need for funding. But Ashton Hayes demonstrated that is possible to get things done without funding. Becoming more formal and professional in general presents another tradeoff. Whether for funding, liability, efficiency, technical specialization or other reasons, it has the effect of limiting who can participate to volunteers or paid professionals with the time and skills (administrative, managerial, technical) to do it. Thus what may have started as an open, inclusive group becomes increasingly specialized and exclusive with the push toward increased productivity and achieving targets. This was most noticeable, though not a major issue, in Forres, the most professionalized initiative which, unlike Eigg and Ashton Hayes, was not incorporated into community governance and where the initiative was somewhat perceived as primarily serving the 'Findhorn' demographic.

Another, more fundamental dilemma of community-based initiatives is whether they can be both mainstream and transformational. Communities are enabled to act yet at the same time constrained in what they can do by strategies of mainstream engagement. All of the cases studied exhibited at least some element of mainstream engagement such as taking an apolitical stance, incorporating into community government, collaborating with local government, and relying on public funding. Without these strategies the initiatives would have struggled to achieve any concrete projects but with them came conditions, explicit and implicit, of bureaucratic compliance and what could be done and said. Although none of the communities were forced into implementing projects unwillingly and there was no indication of proposed projects being denied, projects were selected on what was acceptable to

funders, collaborators, and supporters and it is likely that more radical proposals would have met with resistance. Thus, mainstream engagement allows things to get done but impedes radical change.

Community initiatives can produce real benefits for their communities though they have limited impact in terms of tangible sustainability outcomes when compared to the magnitude of changes required. We tentatively identified a number of strategies that communities may use to improve the effectiveness of interventions and the initiatives that deliver them. But the degree to which transformational outcomes are attained also depends on targeting deeply embedded modes of mobility and consumption and tackling root political and economic drivers. We suggest the case study approach we have taken is a fruitful line of research that should be continued to further determine how community initiatives can be more successful at moving their communities toward sustainability. However, to move beyond incremental interventions, as seen in the four cases described in this article, future research should be oriented towards more transformational outcomes. In particular more research is needed on how to differentiate transformational and incremental solutions and future studies should concentrate on identifying enabling factors and strategies in cases of successful transformational initiatives and interventions.

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